

4CSGC2062: DATA COMMUNICATION AND COMPUTER NETWORKS		
Course Frame Work:		
Credits: L–T–P: 2–0–1-3		Total Credits: 3
Contact Hours/Week: 4	Direct Teaching Hours: 30	Total Contact Hours: 60
Course Learning Objectives: The course aims to introduce the concepts, terminologies and technologies used in data communication and computer networking, familiarize students with division of network functionalities into layers, components required to build different types of networks, functionality of each layer and learn the flow control and congestion control algorithms.		
Course Outcomes (COs): On successful completion of the course, students will be able to:		
CO1	Outline basic concepts in data communications, OSI and TCP/IP protocol stack	L2
CO2	Demonstrate design issues, flow control and error control	L2
CO3	Understand the transfer of data from source to the destination using different protocols and addressing	L2
CO4	Summarize the functions of application layer protocols and how to meet the QoS requirements in networking	L2
CO5	Identify the limits and importance of compression, encoding, sampling, quantization methods	L3
SYLLABUS–THEORY		Hours
Module–I: Data Communication and Networking		06
Introduction; Data communications: components, data representation(ASCII,ISO etc.), direction of data flow(simplex, half duplex, full duplex); Networks: distributed processing, network criteria, physical structure (type of connection, topology), categories of network (LAN, MAN,WAN); Internet: brief history, internet today, the internet, protocols & standards, layered tasks, The OSI model, layers in OSI model, TCP/IP protocol suite, addressing		
Module–II: Design Issues		06
Services to network layer, framing, error control and flow control. Error control: parity bits, hamming codes and CRC. Flow control protocols: unrestricted simplex, stop and wait, sliding window protocol. Medium access control: pure and slotted ALOHA, CSMA, WDMA, CSMA/CD, CSMA/CA		
Module–III: TCP/IP		06
TCP/IP architecture, the Internet protocol, IPv6, UDP, TCP, Internet routing protocols, multicast routing		
Module–IV: Application Layer		06
Application layer overview, Domain Name System (DNS), remote login protocols, e-mail, file transfer and FTP, World Wide Web and HTTP, network management. Overview of QoS, integrated services QoS, differentiated services QoS, virtual private networks		

Module-V: Multimedia Networking	06
Overview of data compression, digital voice and compression, JPEG, MPEG, limits of compression with loss, compression methods without loss, overview of IP telephony, VoIP signaling protocols, real-time media transport protocols, Stream Control Transmission Protocol (SCTP)	
SYLLABUS-LABORATORY	
Simulation using NS2	
1. Point to Point Network: Simulate a three nodes point-to-point network with duplex links between them. Set the queue size vary the bandwidth and find the number of packets dropped. 2. TCP and UDP Traffic Simulation: Simulate a four node point-to-point network, and connect the links as follows: n_0-n_2, n_1-n_2 and n_2-n_3. Apply TCP agent between n_0-n_3 and UDP n_1-n_3. Apply relevant applications over TCP and UDP agents changing the parameter and determine the number of packets by TCP/UDP. 3. Congestion Monitoring: Simulate the transmission of ping message over a network topology consisting of 6 nodes and find the number of packets dropped due to congestion. 4. Error Rate in Ethernet LAN: Simulate an Ethernet LAN using N-nodes (6-10), change error rate and data rate and compare the throughput. 5. Congestion Monitoring in LAN: Simulate an Ethernet LAN using N nodes and set multiple traffic nodes and plot congestion window for different source/destination. 6. Routing Algorithms: Configure RIP/OSPF routing protocols using packet tracer.	
Implementation using C/C++	
7. Routing Algorithms: Write a program for distance vector algorithm to find suitable path for transmission. 8. Error Detection Techniques: Write a program for Hamming code/CRC. 9. Socket Programming: Using TCP/IP sockets, write a client server program to make client sending the file name and the server to send back the contents of the requested file if present. 10. Congestion Control: Write a program for congestion control using leaky bucket algorithm	
EVALUATION SCHEME	

A. Continuous Internal Evaluation (CIE)–Theory:

Components	Internal Assessment Test	Assignment	Total
Max. Marks	20	30	50

Components	Experiment	Viva–Voce	Record Submission	Total
Max. Marks	80	10	10	100

B. Continuous Internal Evaluation (CIE)–Laboratory (Practical)

Note: A student shall obtain a minimum of 50% in each of theory and laboratory components of the course to be eligible to appear for SEE.

C. Semester End Evaluation (SEE): 100 Marks**Question paper pattern:**

1. The question paper shall have FIVE main questions corresponding to the FIVE modules. Internal choices shall be given only in the main questions.
2. Each main question will have TWO full questions carrying TWENTY marks each.
3. A full question may have a maximum of FOUR sub questions, covering the topics under the module.
4. The students will have to answer all FIVE main questions, selecting ONE full question from each module.

Weightage for Final Evaluation		
CIE–Theory	CIE–Laboratory	SEE
0.50	0.25	0.50

BOOKS and REFERENCES

TEXT BOOKS:

- [1] Behrouz A. Forouzan, *Data communication and Networking*, McGraw Hill Education, 4th Edition, 2017, ISBN - 10: 978-0070634145
- [2] A. S. Tanenbaum, *Computer Networks*, 4th Edition, Pearson Education/ PHIR Ajib Mall, ISBN: 9780130661029

REFERENCES:

- [1] Peterson Larry, *Computer Networks: A Systems Approach*, Elsevier, 5th Edition, 2017, ISBN: 978- 9380501932
- [2] William Stallings, *Data and Computer Communication*, Pearson Education, 10th Edition, 2017, ISBN: 978- 8131715000
- [3] James F. Kurose, Keith W. Ross, *Computer Networking: A Top-Down Approach Featuring the Internet*, Pearson Education, 2017, ISBN: 978-9332585492

F. CO-PO-PSO Mapping

CO-PO-PSO Mapping															
CO	PO												PSO		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
1	3	2	3	2	2	-	-	-	-	2	-	3	3	3	2
2	3	3	3	3	3	-	-	-	-	2	-	2	3	3	3
3	3	3	2	3	2	-	-	-	-	2	-	3	3	3	3
4	3	3	2	3	3	-	-	-	-	2	-	2	3	3	2
5	3	3	2	3	3	-	-	-	-	2	-	2	3	3	3